ADVANCED DEEP LEARNING · FINAL REPORT

Facial Emotion Recognition for Medicine and Art

Real-time recognition of seven facial expressions on Vietnamese faces, from data to a local application

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Who needs facial emotion recognition?

MEDICINE

Reading affect when patients cannot tell

- Young children, people after a stroke, with dementia or in intensive care
- Pain and distress show on the face
- Facial behavior studied as a marker of depression
- Hours of recorded therapy and telemedicine sessions

ART

Judging and finding emotional moments

- Does the actor's expression fit the scene?
- Editors choose takes by the strength of a reaction
- The emotional arc of an episode, the response of an audience
- Finding every fearful moment of a 45-minute episode takes hours

Both are done **by eye** today: slow, subjective and rarely measured. A recognizer can do the tedious part; the professional checks the result.

Three problems for a Vietnamese user

01

Faces and data

Public models learn mostly Western web photos or posed laboratory faces, and expressions differ between cultures. Vietnamese TV faces are hardly represented.

02

Privacy

Commercial services send video to the cloud. Face images are personal data under Decree 13/2023/ND-CP; patient video must stay in the clinic.

03

Trust

Microsoft began retiring Azure's emotion recognition in 2022. A professional tool must show what it saw and say when it is unsure.

Task. For every face in a video, a webcam stream or an image, estimate the probabilities of seven expressions (angry, disgust, fear, happy, neutral, sad, surprise), in real time on a laptop, and help the user review long recordings.

Scope: facial expressions, not inner feelings. A research prototype that supports a professional, not a medical device.

Objectives of the final phase

01

Data

A Vietnamese TV face dataset, split by person, at least 1,000 faces per class

02

Models

EmotionCNN, ResNet-18 and MobileNetV2 under one protocol, 3 seeds each

03

Transfer

ImageNet → CK+ → target faces; CK+ tested on people never seen

04

Honest evaluation

Accuracy only on human labels; model-suggested labels give agreement

05

Application

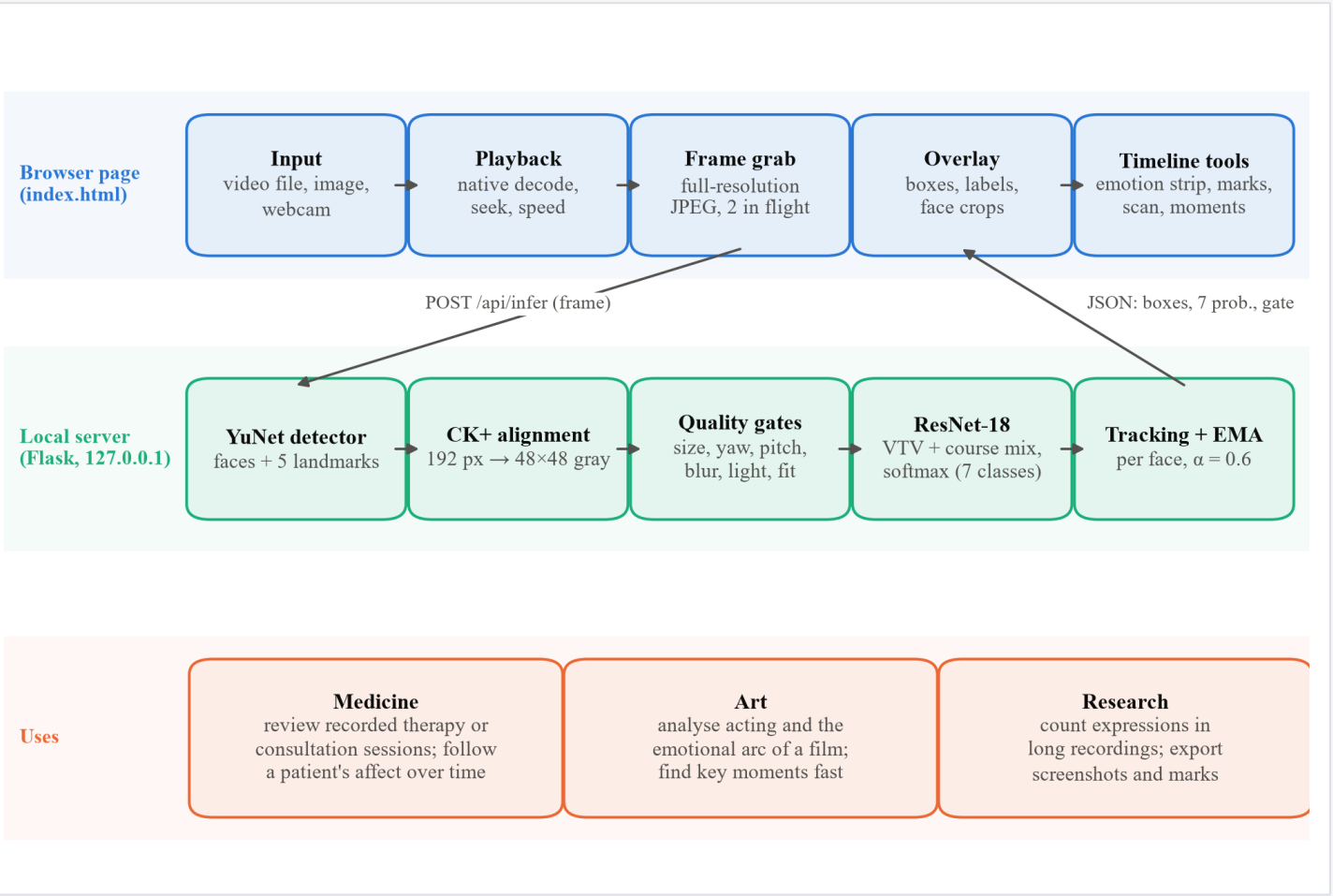
≥ 10 analyzed frames/s at 1080p; seek, jump, own marks, whole-film scan

06

Responsible use

Local only, gray flags outside the training conditions, limits stated

System overview



Browser page

Plays the file natively: seek, speed, own marks, scan. Sends two frames in flight.

Local server

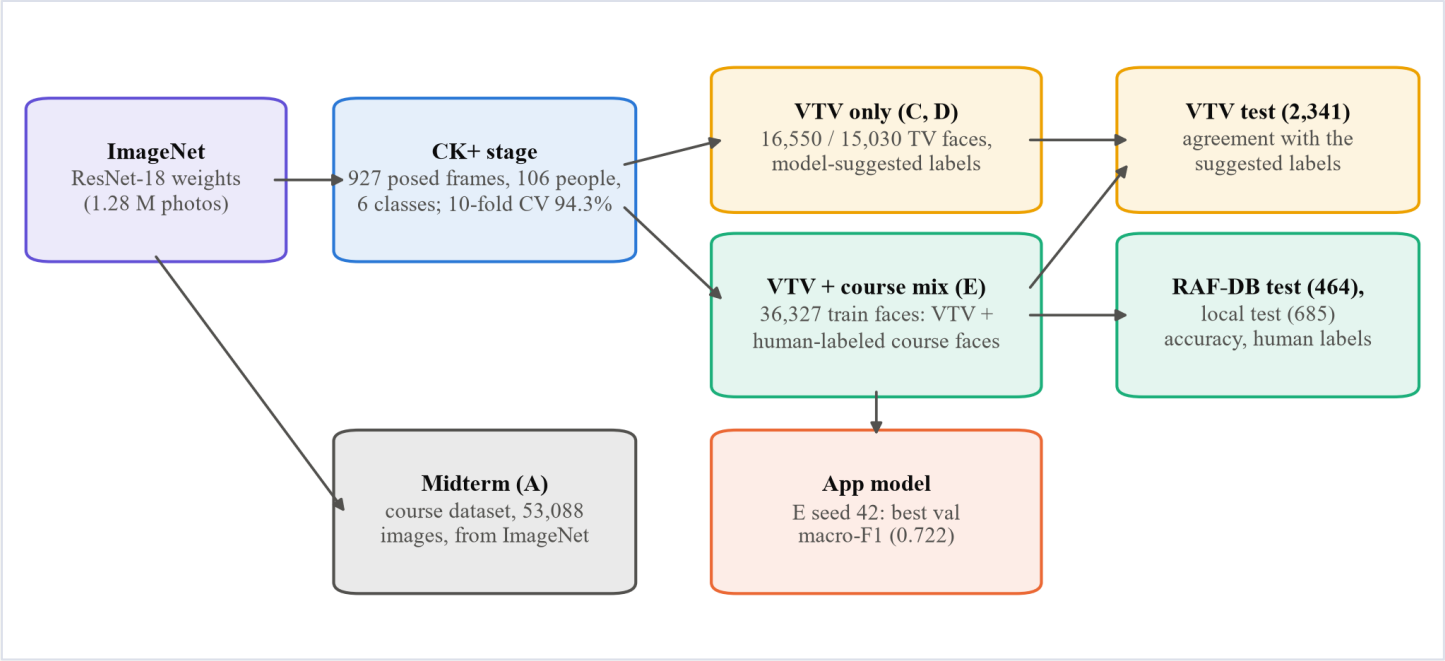
YuNet → CK+ alignment → quality gates → ResNet-18 → tracking and smoothing.

Same pipeline as training

Faces are cut and aligned exactly like the training crops.

Runs on 127.0.0.1: no upload, no account.

Training path: five stages



Transfer twice

ImageNet → CK+ posed expressions → target faces. Only the classifier is re-initialized (6 → 7 outputs).

Two-stage fine-tuning

3 epochs head only, then layer3 / layer4 / head at 5e-5 / 1e-4 / 5e-4; dropout 0.5, weight decay 5e-2.

Same protocol everywhere

Seeds 42, 123 and 2024; model chosen on validation macro-F1; each test set used once.

Stages A and B: ResNet-18 is the most stable

A · Course test set, 1,351 images, human labels

Network	Accuracy (%)	Macro-F1
EmotionCNN	68.02 ± 1.04	0.679
ResNet-18	**68.32 ± 0.97**	**0.683**
MobileNetV2	67.83 ± 0.88	0.678
Ensemble of 9 + flip TTA	71.87	0.717

B · CK+ 10-fold CV by person, 927 images

Network	Accuracy (%)	Macro-F1
EmotionCNN	90.36 ± 1.18	0.867
ResNet-18	**94.28 ± 0.39**	**0.926**
MobileNetV2	93.38 ± 0.53	0.903

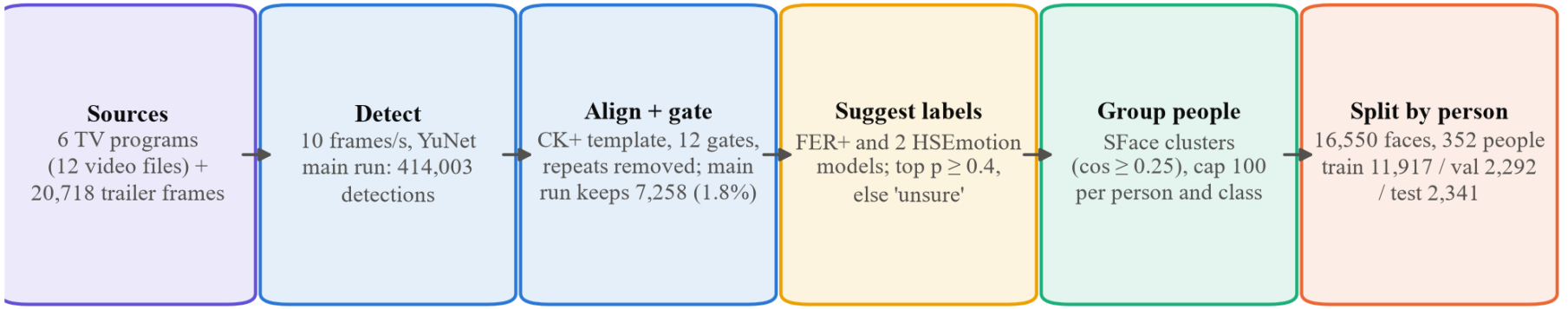
Tied on the course data: less than 0.5 points apart, smaller than the spread between seeds.

Clear posed faces are easy: 94.3% for ResNet-18 on people it never saw.

ResNet-18 varies least over seeds, so it carries stages C–E and the application.

Mean ± std over 3 seeds. CK+ has 6 classes (no neutral).

Building a Vietnamese TV face dataset



All 14 extraction runs kept ~32.7k aligned faces for review; 23,589 were sorted into the 7 classes and 16,550 remain after the per-person cap.

414,003
faces detected in the main extraction run

7,258
passed every quality gate (1.8%)

32.7k
aligned faces kept for review over 14 runs

16,550
faces of 352 people in the final build

VTV dataset: 16,550 faces of 352 people

Class	Train	Val.	Test
angry	788	149	156
disgust	506	85	113
fear	672	111	112
happy	2,816	563	563
neutral	4,629	926	926
sad	1,920	364	364
surprise	586	94	107
total (people)	**11,917 (247)**	**2,292 (54)**	**2,341 (51)**

Train classes under 1,000 faces get 614 augmented copies; validation and test faces are all real. Leak check: 1.3% of test faces have five or more similar training faces.



Aligned 48×48 test faces per suggested class (random sample)

Keeping people and labels honest

ONE PERSON, ONE SPLIT

6% → 0%

of test faces had five or more near-identical training faces; the SFace clustering threshold went from 0.40 to 0.25

86%

of the test set was one comedy performer, until a cap of 100 faces per person and class

SUGGESTED LABELS

FER+ alone → FER+ and two HSEmotion models, on faces prepared like the TV crops

Accuracy on	FER+ alone	Ensemble
CK+	0.659	0.887
RAF-DB	0.548	0.670

Faces with a top probability below 0.4 go to an “unsure” folder. On 48 RAF-DB faces the AI assistant's own eye scored 58%, below the ensemble (71%).

No person has checked these labels yet, so every VTV result is **agreement**, not accuracy.

Stages C and D: high agreement

Network	C: agreement (%)	C: macro-F1	D: agreement (%)	D: macro-F1
EmotionCNN	80.29 ± 1.73	0.731	77.31 ± 3.22	0.614
ResNet-18	**83.64 ± 0.66**	**0.765**	**83.64 ± 0.66**	**0.701**
MobileNetV2	80.21 ± 0.88	0.733	81.41 ± 0.85	0.663

VTV test set (2,341 faces of 51 people) · mean ± std over 3 seeds · agreement with the suggested labels

C: weakest classes

Disgust 57% (taken for angry), fear 66%, surprise 74% (taken for fear): the classes with the fewest faces and the least reliable suggestions.

D: a screen that hurt

The AI assistant removed 1,844 doubtful faces. On 440 shared test faces ResNet-18 fell from 79.7% to 75.8%; surprise from 63–76% to 26–39%.

Lesson

Removing faces emptied the rare classes instead of fixing their labels. C stayed the main VTV dataset.

VTV-ONLY MODEL (D) ON HUMAN LABELS

53.4%

on RAF-DB test faces labeled by people: below the ensemble that made its labels (68.8%)

Surprise recall 11%, disgust 14%. The 83.6% agreement had hidden a weak model: it learned the suggesters' habits from a few hundred rare-class faces.

Stage E: adding human-labeled faces

Source	Train	Val.	Test	Labels
VTV (Vietnamese TV)	11,917 + 614 aug.	2,292	2,341	suggested
local (course)	12,453	2,060	685	human
RAF-DB	8,049	1,396	464	human
AffectNet	2,538	441	153	human
CK+	756	121	45	human
total	**36,327**	**6,310**	**3,688**	

Real course faces (not their augmented copies) are aligned with the CK+ template, like the TV faces, so training and the application see the same crops.

≈ 4,200

training faces per rare class, up from 500–800

unseen

RAF-DB and local test faces: each source keeps its own split

3 × 23 epochs

≈ 29 min per seed on the laptop; best epochs 14–17

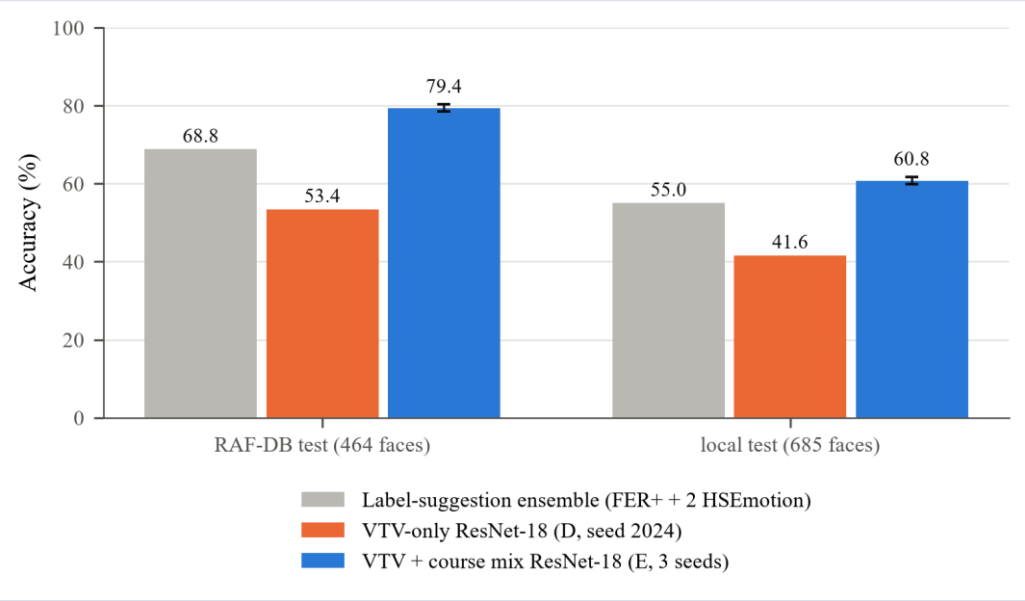
VTV + COURSE MIX (E), 3 SEEDS

79.4%

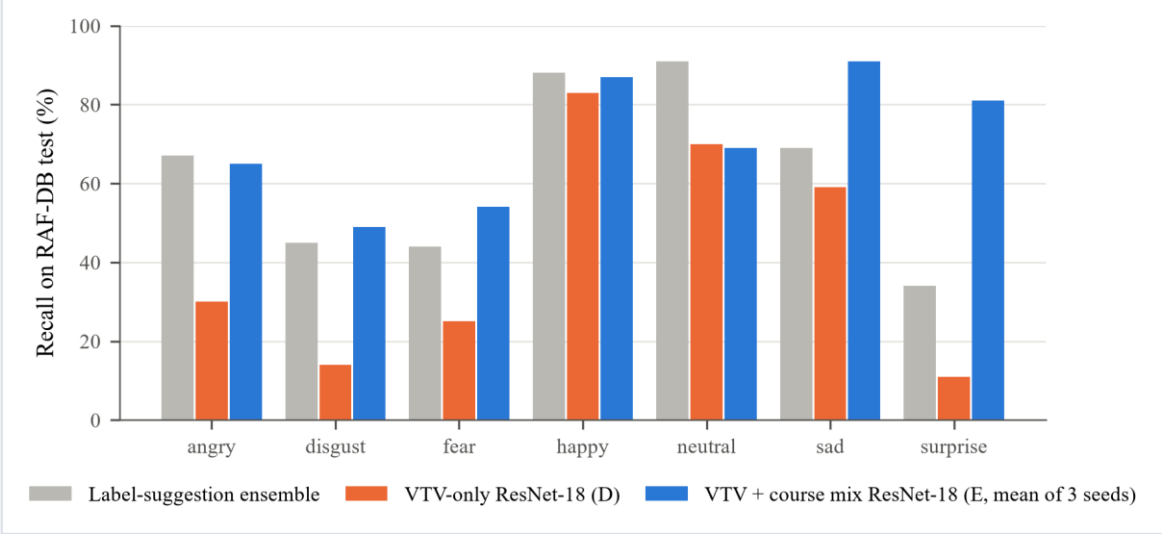
on the same human-labeled RAF-DB test faces (macro-F1 0.713), up from 53.4%

Surprise recall 11% → 81%, disgust 14% → 49%, angry 30% → 65%, sad 59% → 91%. Local test faces 41.6% → 60.8%. Part of the gain is in-distribution: E trained on other RAF-DB and local faces.

Accuracy on human labels



Accuracy on human-labeled test faces (std over 3 seeds)



Recall per class on the 464 RAF-DB test faces

RAF-DB: suggestion ensemble 68.8%, VTV-only 53.4%, mix **79.4%**

Rare classes recover: surprise 11 → 81%, disgust 14 → 49%

Local test faces stay harder: 60.8% (ensemble 55.0%)

All stages at a glance

Stage	Test data	Labels	Accuracy / agreement (%)	Macro-F1
A Midterm	Course test, 1,351	human	68.32 ± 0.97 (ensemble 71.87)	0.683
B CK+ CV	CK+, 927 (10 folds)	human	94.28 ± 0.39	0.926
C CK+ → VTV	VTV test, 2,341	suggested	83.64 ± 0.66	0.765
D CK+ → VTV screened	RAF-DB test, 464	human	53.4 (seed 2024)	0.391
E CK+ → mix	**RAF-DB test, 464**	**human**	**79.4 ± 0.9**	**0.713**
E CK+ → mix	**local test, 685**	**human**	**60.8 ± 0.9**	**0.610**
E CK+ → mix	VTV test of C, 2,341	suggested	82.44 ± 1.24	0.748
Suggestion ensemble	RAF-DB / local test	human	68.8 / 55.0	0.590 / 0.538

Only rows with **human** labels are accuracy; **suggested** rows are agreement with the label-suggestion ensemble.

ResNet-18 unless noted · mean ± std over 3 seeds

The application

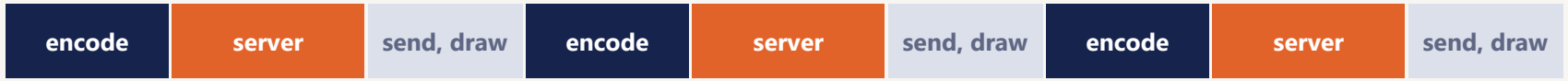


Người Phán Xử, 41-minute episode after a whole-film scan (interface in Vietnamese)

- **Open anything:** video, image or webcam; drag and drop
- **Jump to any time:** 1:23:45, 83:20, 50%, +30
- **Own time marks** with labels, saved per file
- **Whole-film scan:** emotion strip and strongest moments
- **Gray faces:** outside the training conditions
- **48×48 crop** the model actually saw
- **Screenshot** with boxes for a case note or log

Real time on a laptop GPU

First version
one frame at a time · ≈ 8 fps

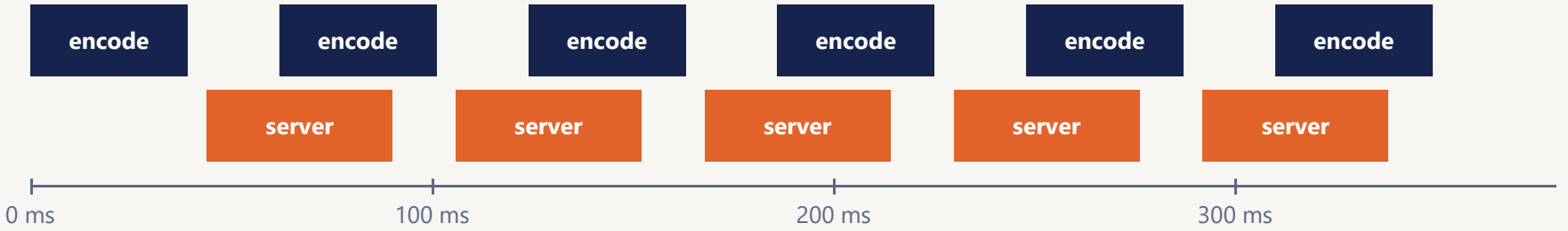


Now: two frames in flight

browser · encodes frame n+1

server · analyzes frame n

≈ 15–17 fps



encode: JPEG of the full 1080p frame in the browser (≈ 40 ms) · server: detect, align, gates, ResNet-18 for every face (≈ 46–49 ms)

15–17 fps
analyzed on 1080p films (13 with many faces)

2.6 ms
per face for ResNet-18 on the GPU (17 ms on the CPU)

70 s
to scan a 41-minute episode at one frame every 10 s

43 MB
model checkpoint; everything runs offline

Built for review work in medicine and art

Function	Medicine	Art
Live view with probabilities	Follow a patient's expressions in a video consultation (with consent)	Feedback for an actor rehearsing in front of a webcam
Whole-file scan + emotion strip	Overview of affect across a recorded therapy session	Emotional arc of an episode; compare scenes or takes
Strongest moments per emotion	Jump to moments of distress for review	Find the strongest reactions for editing or teaching
Own time marks with labels	Clinician's notes at time points	Editor's or director's cue list
Gray faces and 48×48 crop	Shows when not to trust the output	Shows when lighting or angle defeats the model
Screenshot with boxes	Evidence for a case note	Material for a scene analysis

Use on people needs their informed consent. The output supports the professional and never replaces them.

On Vietnamese programs



Frames annotated by the demo (first stage E run)

Close-ups work best

Frontal faces with clear expressions get confident labels.

Gray boxes

Turned, small or blurred faces: outside the training conditions, the label is only a guess.

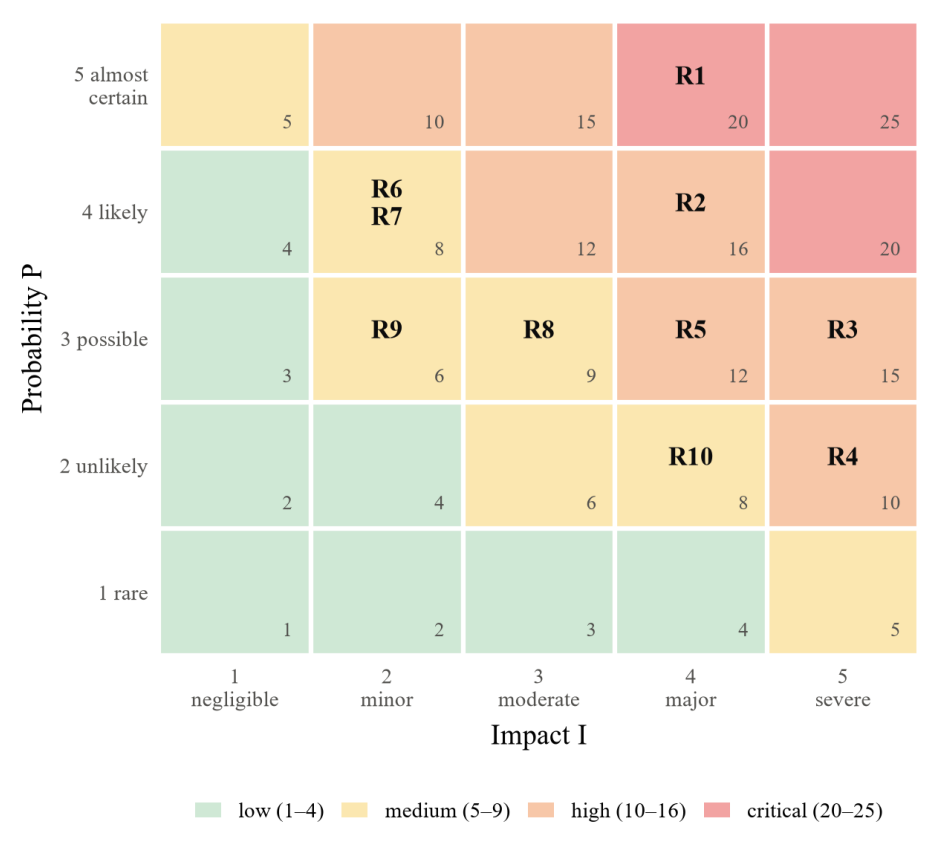
Crowds and stage make-up

Many gray faces in crowd scenes and comedy stage shows.

A fair test is still to come

These programs also supplied training faces; a film outside the dataset is a fairer test.

Risks and how they are handled



ID	Risk	S	Mitigation
R1	Noisy TV labels	20	Human review; human-checked TV test subset; human-labeled faces in training (E)
R2	Domain shift in use	16	Quality gates with gray flags; pilot on consented local data
R3	Misuse or over-reading	15	Probabilities and model crop always shown; no work or school use
R5	Bias	12	Measure accuracy by age, gender and skin tone; add data
R4	Privacy breach	10	Server on 127.0.0.1; no upload; consent for recordings

S = probability × impact (1–5 each): 20–25 critical, 10–16 high

Responsible use

Not a medical device

The output supports a clinician and never replaces one. Clinical use would need IEC 62304, ISO 14971 and a validation study.

Local only

Server bound to 127.0.0.1; no upload, no account. Face images are personal data under Decree 13/2023/ND-CP and Law 91/2025/QH15.

Medicine and art, not monitoring

The EU AI Act bans emotion recognition at work and in schools, except for medical or safety reasons. Those uses are out of scope.

Honest numbers and tools

Accuracy only on human labels. AI assistance (Claude Code) is disclosed. Guided by IEEE 7000 and IEEE 7014-2024 (emulated empathy).

Conclusion and next steps

What we found

- ResNet-18: 68.3% on the course test set, 94.3% on CK+ people it never saw
- 16,550 Vietnamese TV faces of 352 people, split by person
- Human labels exposed a weak model (53%); the mix model reaches 79.4% on RAF-DB
- A local app: any file, any time, own marks, whole-film scan, 13–17 fps

Next

1. Human review of the TV labels, then retrain
2. Grad-CAM: do the models look at eyes, brows and mouth?
3. A faster MobileNetV2 mix model; more real rare-class faces
4. Valence, arousal and intensity outputs
5. Pilots with clinicians and film professionals, with consent

Thank you

Questions?

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